

Problem Set 5

Technology Use and Wellness

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Abstract

For this problem set I examined the relationship between daily technology use and measures of mental health and stress. I produce descriptive visualizations (Part A) and estimate econometric models (Part B) to quantify the relationship.

Research question

How does total daily technology usage relate to mental health scores? Secondary questions:

- How does sleep quality relate to reported stress?
- Does the effect of technology usage on mental health vary with age?

Data

The dataset used is `Tech_Use_Stress_Wellness.csv`, containing over 5,000 observations on technology usage, stress, sleep quality, and wellness factors. This dataset was sourced from Kaggle: [Tech Use and Stress & Wellness Dataset](#).

Key variables include:

- **mental_health_score**: the dependent variable (higher = better mental health).
- **phone_usage_hours**, **laptop_usage_hours**, **tablet_usage_hours**, **tv_usage_hours**: used to compute `total_tech_usage`.
- **sleep_quality**, **stress_level**, **age**, **physical_activity_hours_per_week**: control variables.

Data cleaning steps (in `PS5_mallick.py`):

1. Drop rows with missing values.
2. Create `total_tech_usage` = sum of device usage hours.
3. Categorize `age` into bins (18–25, 26–35, 36–50, 51+).

Part A: Visualizations

Below are the three figures generated by the analysis script. The PNG files exist in the `images/` folder.

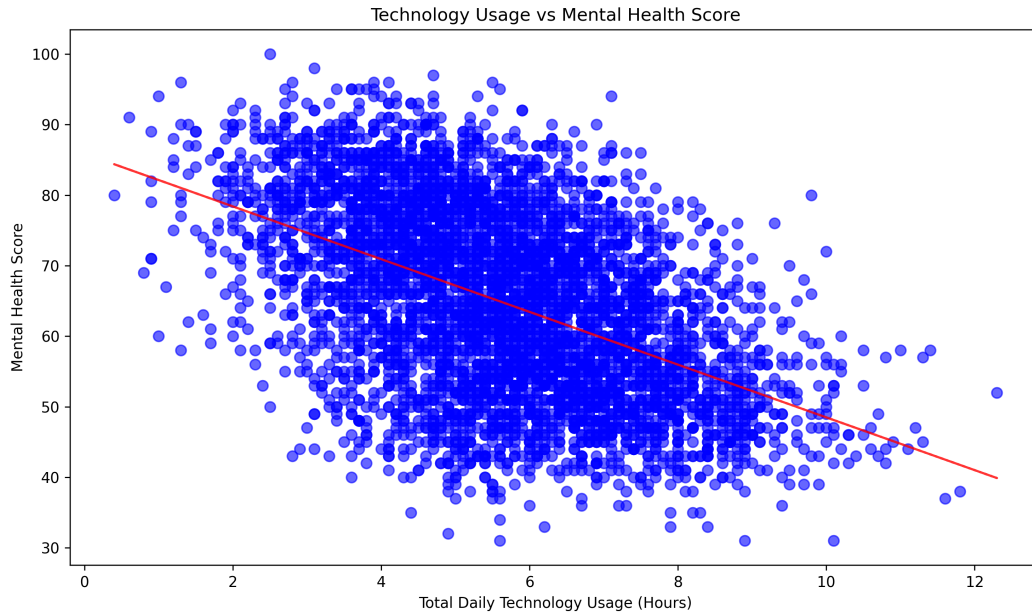


Figure 1: Scatter plot: Total daily technology usage vs. mental health score (trend line).



Figure 2: Bar chart: Average stress level by sleep quality.

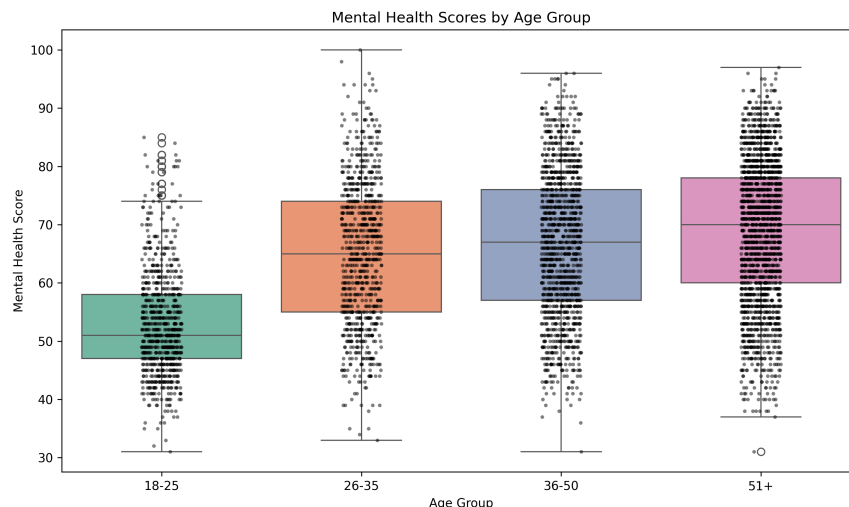


Figure 3: Boxplots: Mental health scores by age group (with raw points).

Part B: Econometric models

I estimate three OLS models:

1. **Basic:** mental_health_score on total_tech_usage.
2. **With controls:** add age, sleep_quality, physical_activity_hours_per_week.
3. **With interaction:** include interaction total_tech_usage $\times$ age.

Regression results

The table below is imported from the results saved by the analysis script. If the table does not appear, confirm that `tables/model_results.tex` exists and contains a valid LaTeX tabular environment.

Model	Tech Usage Coef	P-value	R-squared	Observations
Basic	-3.738	0.000	0.246	5000.000000
With Controls	-1.462	0.000	0.718	5000.000000
With Interaction	-1.469	0.000	0.420	5000.000000

Short interpretation

- **Technology usage:** The coefficient on total technology usage is negative and statistically significant across models, indicating higher usage is associated with lower mental health scores (worse mental health).
- **Controls:** Adding sleep quality and physical activity increases model fit and shows these covariates are important predictors.

- **Interaction:** The interaction with age suggests the effect of technology usage varies with age (the impact per hour changes as age increases).

Unit Tests

I wrote unit tests for the main functions and all passed successfully. The screenshot below shows the pytest results.

```
(.venv) rachi_allick@CODMSBPEDCON4088:/mnt/c/Users/Prachi Mallick/OneDrive - University of South Carolina/Desktop/Econ/CompEcon_Fall25/ProblemSets/Proble
mSet5$ bash run_ps5.sh
1) Running unit tests...
...
4 passed in 85.03s (0:01:25) [100%]
2) Running analysis script...
/mnt/c/Users/Prachi Mallick/OneDrive - University of South Carolina/Desktop/Econ/CompEcon_Fall25/ProblemSets/ProblemSet5/PS5_mallick.py:81: FutureWarnin
g:
Passing 'palette' without assigning 'hue' is deprecated and will be removed in v0.14.0. Assign the 'x' variable to 'hue' and set 'legend=False' for the
same effect.
sns.boxplot(data=df, x="age_group", y="mental_health_score", palette="Set2")
All outputs generated successfully!
Check 'images/' for plots and 'tables/' for regression results.
3) Compiling LaTeX (twice for references)...
pdflatex not found - skipping LaTeX compile
(.venv) rachi_allick@CODMSBPEDCON4088:/mnt/c/Users/Prachi Mallick/OneDrive - University of South Carolina/Desktop/Econ/CompEcon_Fall25/ProblemSets/Proble
mSet5$
```

Figure 4: All 4 unit tests passed successfully (pytest output).

Conclusion

This analysis provides evidence of a negative association between total daily technology use and mental health. Future work could address causality using experimental or panel methods and explore mechanisms (social media use, nighttime usage, etc.).

Files created by code: images/viz1_tech_vs_mental_health.png, images/viz2_sleep_vs_stress.png, images/viz3_mental_health_by_age.png, tables/model_results.csv, tables/model_results.tex.